

Joana Konadu Owusu

Cell: +1 307 761 3714 | **Email:** jowusu1@uwyo.edu | **Website:** <https://joan947.github.io> |

LinkedIn: <https://www.linkedin.com/in/joana-konadu-owusu-8b735b182/> | **GitHub:** <https://github.com/joan947> |

Location: Laramie, Wyoming

Objective/ Summary

I am a mission driven MSc. Computer Science student supervised by Dr. Shivanand Venkanna Sheshappanavar. My research entails LLM and VLMs for Assistive Technologies, Health and Transportation. I am proficient in Machine Learning and Software Development technologies, and I am seeking to launch my career as Research Scientist where I can contribute to meaningful projects and apply my expertise to develop impactful solutions.

Education

Doctor of Philosophy (PhD) –Computer Science
University of Wyoming

Aug 2024 – May 2028
GPA: 4.0

Bachelor of Science (BSc) –Computer Science & Engineering
University of Mines & Technology

Sep 2016 – Aug 2020
CGPA: 3.61

Research Projects

SAM3Count for Zero-Shot Open Vocabulary Counting in Images and Videos (Lead) Nov 2025 – Mar 2026

- Built zero-shot counting model targeting counting failures caused by dense scenes in images and videos
- Designed an ROI and adaptive tiling preprocessor and an identity layer over SAM3 to maintain consistent identities
- Achieved SOTA results across image and video benchmarks tested (over 13% decrease in MAE set).
- Accepted for Oral presentation at CVPR Workshop 2026.

MedQA LLMs (Lead)

Oct 2025 – Nov 2025

- Fine-tuned GPT-2, LLaMA-3.2 1B, and Gemma with LORA on MedQuAD to perform comparative analysis and evaluations with the aim to deduce which LLM is best for Med QA
- Evaluated new models with F1, BERTScore, LLM-as-judge (Ollama) and human evaluations with LLaMA being the highest scorer of F1: 33.38%, BERTScore: 62.55%, human: 4.2/5

Smart Glasses for the Visually Impaired (Lead)

Jan 2025 – Sep 2025

- Engineered an AI-powered smart glasses device, built on a Raspberry Pi, to assist visually impaired in scene analysis and object identification
- Built a cross-platform mobile application to serve as the primary user interface and communication hub to
- Developed a computer vision and LLM model to process sensory data and provide descriptive feedback to the user.

GPT2 from Scratch (Lead)

Sep 2025 – Nov 2026

- Built an end-to-end GPT model with tokenization, attention, chain of thought reasoning and RAG techniques.
- Pretrained on unlabeled data and fine-tuned to follow instructions on instruction–response pairs with logged training loss and accuracy and generation behavior across temperatures.

XPT: Enhanced Point Transformer V3 (Lead)

Jan 2025 – May 2025

- I re-engineered the Point Transformer model used for 3D point cloud understanding to improve its training and inference accuracy (0.5% to 1.5% across Scannet and S3DIS datasets) and speed (40% decrease in speed) showcasing ability to optimize large ML models for production environments.

Estimating Optimal Path Route Finding for Emergency Response (Contributor)

Oct 2024 – Dec 2024

- Applied AI search algorithms (A*, Dijkstra, Bidirectional Search) to design a decision system for emergency route optimization.
- Developed a risk factor model integrated into heuristic functions and implemented route visualization using NetworkX, demonstrating strong algorithmic and data-driven problem-solving skills.
- Worked on visualizing the determined path on the map using NetworkX framework.

Work Experience

Graduate Research Assistant

Aug 2024 – Present

University of Wyoming | WY, USA

Supervisor: Dr.Shivanand Venkanna Sheshappanavar

- Improve and develop VLM (Image, Video and 3D) and LLM models for assistive technologies, healthcare and autonomous driving.
- Work with HPC clusters to incorporate techniques that optimize and enhance efficiency of models and design hardware wearable IoT devices for persons with disabilities
- Assist the REU NSF funded summer program for mentoring undergraduate students to guide them in their research journey.

Information Technology Coordinator

Nov 2021 – July 2024

Ghana Education Service |Kumasi, Ghana

- Administered and maintained the school's database systems and networks while maximizing the information technology tools and systems' efficiency and effectiveness
- Performed Network and System administration tasks on Windows Server and SAP software.
- Reviewed schools' system and network performance to make proactive adjustments based on established IT policies.

Software Developer Intern

June 2018 –Aug 2018

Mobile Web Ghana | Accra, Ghana

- Contributed to a fast-paced startup environment, collaborating with senior engineers in an Agile team to build customized software products using WordPress, Django, React, Node JS and MySQL frameworks.
- Trained beginners (teens to middle aged) in website development using WordPress, Google Sites, HTML, CSS and JavaScript.

Publications

Joana Konadu Owusu, Shivanand Venkanna Sheshappanavar, "SAM3Count for Zero-Shot Open Vocabulary Counting in Images and Videos", (Oral), Computer Vision and Pattern Recognition Workshop Proceedings (CVPR) 2026. [\[PDF\]](#)

Skills & Certifications

Technical Languages: Python, SQL, Java, JavaScript.

Operating Systems: Windows, Linux, Mac OS.

Certifications: LinkedIn Full Stack Software Developer, Microsoft Azure Fundamentals, IBM Enterprise Design Thinking Practitioner

Tools: GitHub, HPC, SLURM, Jupyter Notebook, Pytorch, CUDA, RAG, Node JS, React, MySQL, MongoDB, Microsoft Azure, AWS (SageMaker, Snow Flake, S3, IAM, CloudWatch, EC2, Lambda), ML Ops, CI/CD, REST API, Raspberry PI

Soft Skills: Mission-Driven, Collaborative, Strong Communicator, Eager to Learn, Passion for Growth.

Extracurricular & Leadership Activities

Volunteer Software Development Specialist

Inclusive Tech Group

Nov 2020 – to date

- Member of the core team that plans, host and organize events, training sessions, conferences, fairs and hackathons to advocate for Disability Inclusion in the technology space.
- Collaborate with members of the product development team to develop assistive technologies for persons with disabilities including mobile apps, web platforms and hardware IoT devices.

Volunteer Student AI Expert

University of Wyoming

Nov 2025 – to date

Member of the student committee under the University AI committee, that provide recommendations and guidelines for responsible and ethical use of AI in the classroom and research.